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Splitter and guide vane arrangement for gas turbine engines

Abstract

A section for a gas turbine engine according to an example of the present disclosure includes, among other things, a rotor including a row of blades extending in a radial direction outwardly from a hub. The row of blades deliver flow to a bypass flow path, an intermediate flow path, and a core flow path. A first case surrounds the row of blades to establish the bypass flow path. A first flow splitter divides flow between the bypass flow path and a second duct. An aftmost row of guide vanes extends in the radial direction across the bypass flow path. A second flow splitter radially inboard of the first flow splitter divides flow from the second duct between the intermediate flow path and the core flow path. A bypass port interconnects the intermediate and bypass flow paths.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 16/891,492, filed Jun. 3, 2020, the disclosure of which is incorporated by reference in its entirety herein.

BACKGROUND

(1) This disclosure relates generally to flow distribution for gas turbine engines, and more particularly to arranging guide vanes relative to one or more flow splitters.

(2) Gas turbine engines can include a fan section, a compressor section, a combustor section and a turbine section. The fan section includes a fan having fan blades for compressing a portion of incoming air to produce thrust and also for delivering a portion of air to the compressor section. Air entering the compressor section is compressed and delivered into the combustor section where it is mixed with fuel and ignited to generate a high-speed exhaust gas flow. The high-speed exhaust gas flow expands through the turbine section to drive the compressor section and the fan.

(3) Some fan sections may include guide vanes positioned in a bypass flow path downstream of the fan blades and a flow splitter. The guide vanes may direct the bypass airflow from the fan blades before being ejected from the bypass flow path.

SUMMARY

(4) A section for a gas turbine engine according to an example of the present disclosure includes a rotor including a row of blades extending in a radial direction outwardly from a hub. The hub is rotatable about a longitudinal axis such that the row of blades deliver flow to a bypass flow path, an intermediate flow path, and a core flow path. A first case surrounds the row of blades to establish the bypass flow path. A first flow splitter divides flow between the bypass flow path and a second duct. A row of guide vanes extends in the radial direction across the bypass flow path. A second flow splitter radially inboard of the first flow splitter divides flow from the second duct between the intermediate flow path and the core flow path. A forwardmost edge of the second flow splitter is axially aft of a forwardmost edge of the first flow splitter with respect to the longitudinal axis, and the forwardmost edges of the first and second flow splitters are axially forward of the row of guide vanes with respect to the longitudinal axis. A bypass port interconnects the intermediate and bypass flow paths at a position axially aft of a position of the row of guide vanes relative to the longitudinal axis.

(5) In a further embodiment of any of the foregoing embodiments, the hub is driven by a turbine through a geared architecture.

(6) In a further embodiment of any of the foregoing embodiments, each of the blades extends in the radial direction between a 0% span position at the hub and a 100% span position at a tip to establish a hub-to-tip ratio, and the hub-to-tip ratio is less than or equal to 0.4 measured relative to a forwardmost portion of a leading edge of the blades.

(7) In a further embodiment of any of the foregoing embodiments, a pressure ratio across the blades alone is less than 1.45 at cruise at 0.8 Mach and 35,000 feet.

(8) A further embodiment of any of the foregoing embodiments includes a first duct and a second duct. The first duct establishes the bypass flow path. The second duct branches between the intermediate and core flow paths at the second flow splitter. A first annulus area is established by the first duct at the forwardmost edge of the first flow splitter. A second annulus area is established by the second duct at the forwardmost edge of the first flow splitter. A bypass area ratio is defined as the first annulus area divided by the second annulus area, and the bypass area ratio is greater than or equal to 10.

(9) In a further embodiment of any of the foregoing embodiments, each of the blades extends in the radial direction between a 0% span position at the hub and a 100% span position at a tip. The

forwardmost edge of the first flow splitter is situated in the radial direction at a first splitter position. The first splitter position is radially aligned with or radially inward of a 25% span position of the blades. The forwardmost edge of the second flow splitter is situated in the radial direction at a second splitter position, and the second splitter position is radially aligned with or radially outward of a 5% span position of the blades.

(10) In a further embodiment of any of the foregoing embodiments, each of the blades extends in the radial direction between the 0% span position and the 100% span position to establish a hub-to-tip ratio, and the hub-to-tip ratio is between 0.16-0.36. The hub-to-tip ratio measured relative to a forwardmost portion of a leading edge of the blades.

(11) In a further embodiment of any of the foregoing embodiments, the leading edges of the blades at the 100% span position is established along a first reference plane, and leading edges of the guide vanes at a 100% span position are established along a second reference plane. The first and second reference planes are perpendicular to the longitudinal axis. A first axial length is established between the first and second reference planes. A tip radius of the blades is established between the tip and the longitudinal axis, and a ratio of the first axial length divided by the tip radius is greater than or equal to 0.5.

(12) In a further embodiment of any of the foregoing embodiments, the row of blades establishes a blade quantity (BQ), the blade quantity (BQ) is at least 12 but not more than 20 blades, the row of guide vanes establishes a vane quantity (VQ), and the vane quantity (VQ) is at least 20 but not more than 40 guide vanes.

(13) In a further embodiment of any of the foregoing embodiments, the ratio of VQ/BQ is between 2.0 and 2.6.

(14) In a further embodiment of any of the foregoing embodiments, the first splitter position radially is aligned with or radially outward of a 10% span position of the blades, and the second splitter position is radially aligned with or radially inward of a 15% span position of the blades. A second axial length is established between the first and second splitter positions relative to the longitudinal axis, and a ratio of the second axial length divided by the first axial length is between 0.1 and 0.3.

(15) A gas turbine engine according to an example of the present disclosure includes a fan section including a fan having a row of blades extending in a radial direction between a 0% span position at a hub and a 100% span position at a tip. The hub is rotatable about an engine longitudinal axis such that the row of blades deliver flow to a bypass flow path, an intermediate flow path and a core flow path. A compressor section establishes the core flow path. A turbine section drives the fan section and the compressor section. A fan case includes a bypass duct surrounding the row of blades to establish the bypass flow path. A housing includes a first flow splitter that divides flow between the bypass flow path and a second duct. A row of guide vanes in the bypass duct extend in the radial direction across the bypass flow path. An engine case includes a second flow splitter radially inboard of the first flow splitter and divides flow from the second duct between the intermediate flow path and the core flow path. A bypass port interconnects the intermediate and bypass flow paths at a position downstream of the row of guide vanes. Each of the blades extends in the radial direction between the 0% span position and 100% span position to establish a hub-to-tip ratio, and the hub-to-tip ratio is less than or equal to measured relative to a forwardmost portion of a leading edge of the blades.

(16) In a further embodiment of any of the foregoing embodiments, the turbine section includes a fan drive turbine, and a geared architecture drives the fan at a different speed than a speed of the fan drive turbine.

(17) In a further embodiment of any of the foregoing embodiments, a pressure ratio across the blades alone is less than 1.45 at cruise at 0.8 Mach and 35,000 feet.

(18) In a further embodiment of any of the foregoing embodiments, a forwardmost edge of the second flow splitter is axially aft of a forwardmost edge of the first flow splitter with respect to the

engine longitudinal axis, and the forwardmost edges of the first and second flow splitters are axially forward of the row of guide vanes with respect to the engine longitudinal axis.

(19) A further embodiment of any of the foregoing embodiments includes one or more blocker doors situated in the intermediate flow path downstream of the bypass port. The one or more blocker doors are moveable to selectively communicate flow in the intermediate flow path to a vent port.

(20) A further embodiment of any of the foregoing embodiments includes a thrust reverser including a cascade that selectively communicates airflow from the bypass duct. The cascade extends axially aft of both the row of guide vanes and the bypass port with respect to the engine longitudinal axis.

(21) A method of operation for a gas turbine engine according to an example of the present disclosure includes driving a rotor having a hub and a row of blades to deliver airflow to a bypass flow path, and an intermediate flow path and a core flow path. A first duct surrounds the row of blades to establish the bypass flow path. A first flow splitter divides airflow between the bypass flow path and the intermediate flow path. A row of guide vanes extend across the bypass flow path. A second flow splitter divides airflow between the intermediate flow path and the core flow path. The method includes communicating the airflow through a bypass port that interconnects the intermediate flow path to the bypass flow path at a position downstream of the row of guide vanes. Each of the blades extends in a radial direction between a 0% span position at the hub and a 100% span position at a tip to establish a hub-to-tip ratio, and the hub-to-tip ratio is less than or equal to 0.4 measured relative to a forwardmost portion of a leading edge of the blades.

(22) In a further embodiment of any of the foregoing embodiments, the method includes selectively moving one or more blocker doors to communicate airflow in the intermediate flow path to a vent port, the vent port established downstream of the bypass port.

(23) In a further embodiment of any of the foregoing embodiments, the driving step including driving the rotor through a geared architecture at a different speed than a turbine, the hub-to-tip ratio is between 0.16-0.36, a pressure ratio across the blades alone is less than 1.45 at cruise at 0.8 Mach and 35,000 feet, and a first annulus area is established at the forwardmost edge of the first flow splitter. A second annulus area is established at the forwardmost edge of the first flow splitter. A bypass area ratio is defined as the first annulus area divided by the second annulus area, and the bypass area ratio is greater than or equal to 10.

(24) The various features and advantages of this disclosure will become apparent to those skilled in the art from the following detailed description. The drawings that accompany the detailed description can be briefly described as follows.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 illustrates a gas turbine engine.

(2) FIG. 2A illustrates an example arrangement of blades and guide vanes at a span position.

(3) FIG. 2B illustrates an example arrangement of blades and guide vanes at another span position.

(4) FIG. 3 is a schematic view of airfoil span positions for a blade.

(5) FIG. 4 is a schematic view of airfoil span positions for a guide vane.

(6) FIG. 5 illustrates a gas turbine engine according to another example including blocker doors in a closed position.

(7) FIG. 6 illustrates the blocker doors of FIG. 5 in an opened position.

(8) FIG. 7A is a partial cross section view of a thrust reverser and a variable area nozzle in stowed positions.

(9) FIG. 7B is a partial cross section view of the thrust reverser of FIG. 7A in the stowed position

and the variable area nozzle of FIG. 7A in a deployed position.

(10) FIG. 7C is a partial cross section view of the thrust reverser and the variable area nozzle of FIG. 7A in deployed positions.

(11) FIG. 8A illustrates a section of the engine of FIG. 5.

(12) FIG. 8B illustrates a sectional view taken along line 8B-8B of FIG. 8A.

(13) FIG. 8C illustrates an exemplary guide vane.

(14) FIG. 9A illustrates an example plot of bypass area ratio with respect to span position.

(15) FIG. 9B illustrates selected portions of the plot of FIG. 9A.

(16) FIG. 10A illustrates an example plot of fan pressure ratio with respect to span position.

(17) FIG. 10B illustrates selected portions of the plot of FIG. 10A.

(18) FIG. 11 illustrates another example plot of fan pressure ratio with respect to span position.

(19) FIG. 12 illustrates exemplary engine modes.

(20) Like reference numbers and designations in the various drawings indicate like elements.

DETAILED DESCRIPTION

(21) FIG. 1 schematically illustrates a gas turbine engine 20. The gas turbine engine 20 is disclosed herein as a two-spool turbopfan that generally incorporates a fan section 22, a compressor section 24, a combustor section 26 and a turbine section 28. The fan section 22 drives air along a bypass flow path B in a bypass duct defined within a housing 15 such as a fan case or nacelle, and also drives air along a core flow path C for compression and communication into the combustor section 26 then expansion through the turbine section 28. Although depicted as a two-spool turbopfan gas turbine engine in the disclosed non-limiting embodiment, it should be understood that the concepts described herein are not limited to use with two-spool turbopfans as the teachings may be applied to other types of turbine engines including three-spool architectures.

(22) The exemplary engine 20 generally includes a low speed spool 30 and a high speed spool 32 mounted for rotation about an engine central longitudinal axis A relative to an engine static structure 36 via several bearing systems 38. It should be understood that various bearing systems 38 at various locations may alternatively or additionally be provided, and the location of bearing systems 38 may be varied as appropriate to the application.

(23) The low speed spool 30 generally includes an inner shaft 40 that interconnects, a first (or low) pressure compressor 44 and a first (or low) pressure turbine 46. The inner shaft 40 is connected to the fan 42 through a speed change mechanism, which in exemplary gas turbine engine 20 is illustrated as a geared architecture 48 to drive a fan 42 at a lower speed than the low speed spool 30. The high speed spool 32 includes an outer shaft 50 that interconnects a second (or high) pressure compressor 52 and a second (or high) pressure turbine 54. A combustor 56 is arranged in exemplary gas turbine 20 between the high pressure compressor 52 and the high pressure turbine 54. A mid-turbine frame 57 of the engine static structure 36 may be arranged generally between the high pressure turbine 54 and the low pressure turbine 46. The mid-turbine frame 57 further supports bearing systems 38 in the turbine section 28. The inner shaft 40 and the outer shaft 50 are concentric and rotate via bearing systems 38 about the engine central longitudinal axis A which is collinear with their longitudinal axes.

(24) The core airflow is compressed by the low pressure compressor 44 then the high pressure compressor 52, mixed and burned with fuel in the combustor 56, then expanded through the high pressure turbine 54 and low pressure turbine 46. The mid-turbine frame 57 includes airfoils 59 which are in the core airflow path C. The turbines 46, 54 rotationally drive the respective low speed spool 30 and high speed spool 32 in response to the expansion. It will be appreciated that each of the positions of the fan section 22, compressor section 24, combustor section 26, turbine section 28, and fan drive gear system 48 may be varied. For example, gear system 48 may be located aft of the low pressure compressor, or aft of the combustor section 26 or even aft of turbine section 28, and fan 42 may be positioned forward or aft of the location of gear system 48.

(25) The engine 20 in one example is a high-bypass geared aircraft engine. In a further example,

the engine **20** bypass ratio is greater than about six (6), with an example embodiment being greater than about ten (10), the geared architecture **48** is an epicyclic gear train, such as a planetary gear system or other gear system, with a gear reduction ratio of greater than about 2.3 and the low pressure turbine **46** has a pressure ratio that is greater than about five. In one disclosed embodiment, the engine **20** bypass ratio is greater than about ten (10:1), the fan diameter is significantly larger than that of the low pressure compressor **44**, and the low pressure turbine **46** has a pressure ratio that is greater than about five 5:1. Low pressure turbine **46** pressure ratio is pressure measured prior to inlet of low pressure turbine **46** as related to the pressure at the outlet of the low pressure turbine **46** prior to an exhaust nozzle. The geared architecture **48** may be an epicycle gear train, such as a planetary gear system or other gear system, with a gear reduction ratio of greater than about 2.3:1 and less than about 5:1. It should be understood, however, that the above parameters are only exemplary of one embodiment of a geared architecture engine and that the present invention is applicable to other gas turbine engines including direct drive turbofans.

(26) A significant amount of thrust is provided by the bypass flow B due to the high bypass ratio. The fan section **22** of the engine **20** is designed for a particular flight condition—typically cruise at about 0.8 Mach and about 35,000 feet (10,668 meters). The flight condition of 0.8 Mach and 35,000 ft (10,668 meters), with the engine at its best fuel consumption —also known as “bucket cruise Thrust Specific Fuel Consumption (‘TSFCT’)”—is the industry standard parameter of lbm of fuel being burned divided by lbf of thrust the engine produces at that minimum point. “Low fan pressure ratio” is the pressure ratio across the fan blade alone, without a Fan Exit Guide Vane (“FEGV”) system. The low fan pressure ratio as disclosed herein according to one non-limiting embodiment is less than about 1.45. “Low corrected fan tip speed” is the actual fan tip speed in ft/sec divided by an industry standard temperature correction of $[(T_{\text{ram}} \text{ } ^\circ \text{R}) / (518.7^\circ \text{R})]^{\text{sup.}0.5}$. The “Low corrected fan tip speed” as disclosed herein according to one non-limiting embodiment is less than about 1150 ft/second (350.5 meters/second).

(27) The fan **42** includes at least one row **60** of airfoils or blades **62** that are circumferentially distributed about, and are supported by, a rotatable hub **64**. The row **60** of blades **62** can extend in a generally spanwise or radial direction R outwardly from the hub **64**. The radial direction R can be perpendicular to the engine axis A. The hub **64** is rotatable in a clockwise or counter-clockwise direction RR (FIG. 4) about the engine axis A such that the blades **62** deliver airflow to the bypass flow path B and core flow path C in operation. In the illustrative example of FIG. 1, the hub **64** is driven by the low pressure (or fan drive) turbine **46** through the geared architecture **48**. A spinner or nosecone **65** is supported relative to the hub **64** to provide an aerodynamic inner flow path into the fan section **22**.

(28) The fan section **22** includes at least one row **66** of turning or fan exit guide vanes **68**. The guide vanes **68** are positioned in the bypass flow path B axially aft of the row **60** of blades **62** relative to the engine axis A. The row **66** of guide vanes **68** extend in the radial direction R across the bypass flow path B between the fan (e.g., first) case **15** and a core engine (e.g., second case) **17**. The fan case **15** includes a bypass (e.g., first) duct **19** surrounding the row **60** of blades **62** and the row **66** of guide vanes **68** to establish the bypass flow path B. The engine case **17** is dimensioned to extend about and along the engine axis A to at least partially surround and house the compressors **44**, **52**, combustor **56** and turbines **46**, **54**. The fan case **15** and engine case **17** establish a portion of the engine static structure **36**.

(29) The engine case **17** includes a flow splitter **70** situated downstream and axially aft of the row **60** of blades **62** relative to the engine axis A. The flow splitter **70** can have a generally V-shaped cross sectional geometry and is arranged to divide airflow communicated from the row **60** of blades **62** between the bypass flow path B and a second duct **21** establishing the core flow path C. The second duct **21** can serve as an entrance or inlet to the compressor section **24** and is bounded by the flow splitter **70**.

(30) Referring to FIGS. 2-3, with continuing reference to FIG. 1, each of the blades **62** includes an

airfoil body **62A** that extends in the radial direction **R** from the hub **64** (FIG. **1**) between a root **62R** coupled to the hub **64** and a tip **62T**. Each airfoil body **62A** extends axially in an axial or chordwise direction **H** between a leading edge **62LE** and a trailing edge **62TE**, and extends circumferentially in a thickness or circumferential direction **T** between a pressure side **62P** and a suction side **62S**. The directions **R**, **H**, **T** are perpendicular to each other. The chordwise direction **H** can be substantially parallel or transverse to the engine axis **A**. The radial direction **R** can have a major component that extends generally from or toward an axis of rotation of the blades **62**, which in the illustrated example coincides with the engine axis **A**. It should be understood that the radial direction **R** can include a minor component in the axial and/or circumferential directions **H**, **T** such that the blades **62** have a predefined amount of sweep and/or lean, for example.

(31) The airfoil body **62A** has an exterior airfoil surface **62E** providing a contour that extends in the chordwise direction **H** between the leading and trailing edges **62LE**, **62TE**. The exterior airfoil surface **62E** generates lift based upon its geometry and directs flow along the core flow path **C** and bypass flow path **B**. Each blade **62** can be constructed from a composite material, a metallic material such as steel, an aluminum or titanium alloy, or a combination of one or more of these. Abrasion-resistant coatings or other protective coatings may be applied to the blade **62**.

(32) Referring to FIGS. **2** and **4**, with continuing reference to FIGS. **1** and **3**, each of the guide vanes **68** includes an airfoil body **68A** that extends in the radial direction **R** between inner and outer surfaces **19A**, **19B** of the duct **19**, axially in the chordwise direction **H** between a leading edge **68LE** and a trailing edge **68TE**, and circumferentially in the thickness direction **T** between a pressure side **68P** and a suction side **68S**. The inner surfaces **19A** of the first duct **19** can be provided by the engine case **17** at a location downstream of the flow splitter as illustrated in FIG. **1**.

(33) The airfoil body **68A** has an exterior airfoil surface **68E** providing a contour that extends in the chordwise direction **H** between the leading and trailing edges **68LE**, **68TE**. The exterior airfoil surface **68E** can be contoured to direct airflow **F** (FIGS. **2A**, **2B**) compressed by the blades **62** through a gas path such as the bypass flow path **B**. The guide vanes **68** can be constructed from a composite material, a metallic material such as steel, an aluminum or titanium alloy, or a combination of one or more of these. The guide vanes **68** can serve as a structural component to transfer loads between the fan case **15** and the engine case **17** or another portion of the engine static structure **36**. The fan section **22** of FIG. **1** includes a single stage fan **42** having only one row **60** of blades **62** and only one row **66** of guide vanes **68**. In other examples, the fan section **22** has more than one row of blades **62** and/or guide vanes **68** configured with respect to any of the quantities disclosed herein.

(34) FIGS. **3-4** schematically illustrate span positions of a blade **62** and guide vane **68**, respectively. Span positions are schematically illustrated from 0% to 100% in 25% increments, for example, to define a plurality of sections **62C** of the blade **62** and a plurality of sections **68C** of the guide vane **68**. Each section **62C**, **68C** at a given span position is provided by a conical cut that corresponds to the shape of segments of the respective gas path, such as the bypass flowpath **B** or core flow path **C**, as shown by the large dashed lines.

(35) In the case of a blade **62** with an integral platform, the 0% span position (or zero span) corresponds to the generally radially innermost location where airfoil body **62A** meets the fillet joining the airfoil body **62A** to an adjacent platform **69** (see also FIG. **1**). In the case of a blade **62** without an integral platform, the 0% span position corresponds to the generally radially innermost location where the discrete platform **69** meets the exterior blade surface **62E** of the airfoil body **62A**. A 100% span position (or full span) corresponds to a section **62C** of the blade **62** at the tip **62T**. The 50% position (or midspan) corresponds to a generally radial position halfway between the 0% and 100% span positions.

(36) The 0% span position of the vane **68** can correspond to the generally radially innermost location where the exterior surface **68E** of the airfoil body **68A** meets the inner surfaces **19A** of the duct **19**. The 100% span position can correspond to the generally radially outermost location where

the exterior surface **68E** of the airfoil body **68A** meets the outer surfaces **19B** of the duct **19**. The 50% span position (or midspan) corresponds to a generally radial position halfway between the 0% and 100% span positions. As illustrated in FIGS. **2A** and **2B**, each blade **62** is sectioned at a first generally radial position between the root **62R** and the tip **62T**, and each vane **68** is sectioned at a second generally radial position between the inner and outer surfaces **19A**, **19B** of the duct **19**. The first and second radial positions can be the same (e.g., both at 25%, 50% or 100% span) or can differ (e.g., one at 50% and the other at 100% span).

(37) Airfoil geometric shapes, stacking offsets, chord profiles, stagger angles, axial sweep and dihedral angles, tangential lean angles, bow, and/or other three-dimensional geometries, among other associated features, can be incorporated individually or collectively into the blades **62** and/or guide vanes **68** to improve various characteristics such as aerodynamic efficiency, structural integrity, and vibration mitigation.

(38) Referring to FIG. **2A**, the blades **62** and guide vanes **68** can be arranged at various orientations relative to the engine axis **A**. Each blade **62** establishes a chord dimension **BCD**. Each vane **68** establishes a chord dimension **VCD**. The chord dimension **BCD/VCD** is a length between the respective leading and trailing edges **62LE/68LE**, **62TE/68TE**. The chord dimension **BCD/VCD** may vary along the span of the airfoil body **62A/68A**. The chord **BCD/VCD** forms a respective stagger angle α/β relative to the chordwise direction **H** or a plane parallel to the engine axis **A**.

(39) The stagger angles α , β can vary along the airfoil span to define a twist. For example, the blades **62** and guide vanes **68** in FIG. **2A** may be at a first span position, such as at 0% span, and the blades **62** and guide vanes **68** in FIG. **2B** may be at a second, different span position, such as at 100% span. Referring to FIG. **2B**, the stagger angle α of the blades **62** is larger at 100% span than the stagger angle α of the blades **62** corresponding to the span position of FIG. **2A**. In examples, the stagger angle α of the blade **62** is greater than or equal to about 5 degrees at 0% span, or more narrowly less than or equal to about 40 degrees, such as between about 10 and 30 degrees at 0% span. In examples, the stagger angle α is less than or equal to about 70 degrees at 100% span, or more narrowly less than or equal to about 60 degrees, such as between about 30 and 50 degrees at 100% span.

(40) The stagger angle β of the vane **68** can be substantially the same from 0% span to 100% span or can differ. In examples, the stagger angle β is greater than or equal about 10 degrees at 0% span, or more narrowly less than or equal to about 60 degrees, such as between about 20 and 40 degrees at 0% span. In examples, the stagger angle β is less than or equal about degrees at 100% span, or more narrowly less than or equal to about 60 degrees, such as between about 30 and 50 degrees at 100% span.

(41) Each of the blades **62** can be cambered. A mean camber line **62M** bisects the airfoil body **62A** between the leading and trailing edges **62LE**, **62TE** of the blade **62**. A leading edge metal angle $\alpha1^*$ at the leading edge **62LE** and a trailing edge metal angle $\alpha2^*$ at the trailing edge **62TE** are established with respect to the mean camber line **62M** and the chordwise direction **H**. The tip **62T** of the blade **62** can rotate at a relatively faster speed, and the stagger of the blade **62** can be aligned with a relative flow direction **F'** to achieve an optimum incidence of the flow **F** relative to the leading edge metal angle $\alpha1^*$ of the blade **62**. A camber $\kappa1$ of the blade **62** is equal to a difference between the leading edge metal angle $\alpha1^*$ and the trailing edge metal angle $\alpha2^*$. In the illustrative example of FIGS. **2A-2B**, the camber $\kappa1$ of the blade **62** can be substantially less at 100% span (FIG. **2B**) than the camber $\kappa1$ of the blade **62** at less than 100% span (FIG. **2A**). The camber $\kappa1$ of the airfoil body **62A** contour that extends in the chordwise direction **H** between the leading and trailing edges **62LE**, **62TE** of the blade **62** can be less than or equal to about 80 degrees at 0% span, or more narrowly less than or equal to about 70 degrees, such as between about 40 and 60 degrees at 0% span. The camber $\kappa1$ can be greater than or equal to about 5 degrees at 100% span, or more narrowly less than or equal to about 50 degrees, such as between about 10 and 40 degrees at 100% span.

(42) Each of the vanes **68** can be cambered. A mean camber line **68M** bisects the airfoil body **68A** between the leading and trailing edges **68LE**, **68TE** of the vane **68**. A leading edge metal angle $\beta 1^*$ at the leading edge **68LE** and a trailing edge metal angle $\beta 2^*$ at the trailing edge **68TE** are established with respect to the mean camber line **68M** and the chordwise direction H. A camber $\kappa 2$ of the vane **68** is equal to a difference between the leading edge metal angle $\beta 1^*$ and the trailing edge metal angle $\beta 2^*$. The camber $\kappa 2$ of the airfoil body **68A** contour that extends in the chordwise direction H between the leading and trailing edges **68LE**, **68TE** of the vane **68** can be greater than or equal to about 5 degrees at 0% span, or more narrowly less than or equal to about 50 degrees, such as between about 10 and 40 degrees at 0% span. The camber $\kappa 2$ can be less than or equal to about 80 degrees at 100% span, or more narrowly less than or equal to about 70 degrees, such as between about 40 and 60 degrees at 100% span. For the purposes of this disclosure, the terms “about” and “approximately” mean $\pm 3\%$ of the stated value unless otherwise indicated.

(43) FIG. 5 illustrates a gas turbine engine **120** according to another example. In this disclosure, like reference numerals designate like elements where appropriate and reference numerals with the addition of one-hundred or multiples thereof designate modified elements that are understood to incorporate the same features and benefits of the corresponding original elements. The engine **120** includes a fan section **122** including a fan (or rotor) **142**, a compressor section **124**, a combustor section **126** and a turbine section **128**. The compressor, combustor and turbine sections **124**, **126**, **128** establish respective portions of the core flow path C. The turbine section **128** drives the fan section **122** and the compressor section **124**. The compressor section **124** includes a low pressure compressor **144** and a high pressure compressor **152**. The turbine section **128** includes a low pressure (or fan drive) turbine **146** and a high pressure turbine **154**.

(44) The fan **142** includes at least one row **160** of blades **162** carried by and mounted to a rotatable hub **164**. In examples, the blades **162** are incorporated into a fixed-pitch arrangement in which the blades **162** have a fixed stagger angle α . In other examples, the blades **162** are incorporated into a variable-pitch arrangement in which the stagger angle α varies in response to rotation of the blades **162** about a respective blade axis that extends between the blade root **162R** and tip **162T**.

(45) The engine **120** can include a geared architecture **148** that drives the fan **142** at a different speed than a speed of the fan drive turbine **146** in operation. In examples, the geared architecture **148** is an epicyclic gear system such as a planetary gear system having a fixed ring gear and a rotatable carrier, or such as a star gear system having a fixed carrier and a rotatable ring gear. The geared architecture **48** can establish a gear reduction ratio of greater than about 2.3 or 2.6, or more narrowly less than about 4.0 or 5.0, for example. In other examples, the geared architecture **148** is omitted such that the fan drive turbine **146** directly drives the fan **142** at a common speed and in a common direction.

(46) The fan section **122** includes a fan (e.g., first) case **115** including a bypass (e.g., first) duct **119** surrounding the row **160** of blades **162** to establish a bypass (e.g., first) flow path B. The engine **120** includes an engine (e.g., second) case **117** that establishes a core (e.g., second) flow path C.

(47) The engine **120** can include a nacelle **123** and core cowling (or housing) **125**. The nacelle **123** is mechanically attached or otherwise secured to the fan case **115** and establishes an engine inlet **1231**. The engine inlet **1231** conveys airflow F to the fan **142** in operation. The core cowling **125** is mechanically attached or otherwise secured to the engine case **117**. The core cowling **125** at least partially surrounds the engine case **117**. The bypass flow path B terminates at a bypass exit (or nozzle) **131** established by a trailing edge of the nacelle **123**.

(48) At least one row **166** of turning or fan exit guide vanes **168** are situated in the bypass duct **119**. The guide vanes **168** extend in a radial direction R across the bypass flow path B. Each guide vane **168** extends in the radial direction R between a 0% span position along an inner surface **119A** and a 100% span position along an outer surface **119B** of the bypass duct **119**.

(49) An airfoil body **168A** of the guide vane **168** extends along a longitudinal axis LA, as illustrated in FIG. 8C. A projection of the longitudinal axis LA intersects the engine axis A to

establish a vane angle ϕ . The vane angle ϕ can be approximately 90 degrees such that the guide vane is substantially perpendicular to the engine axis A, as illustrated by the guide vane **68** of FIG. **1**. In examples, the vane angle ϕ is less 90 degrees such that the guide vane has a component that extends aftwards in the axial direction from 0% to 100% span with respect to the engine axis A, as illustrated by the guide vane **168** of FIG. **8A**. The vane angle ϕ can be greater than or equal to 15 degrees, or more narrowly between 30 and 60 degrees, for example.

(50) Various quantities of blades **162** and guide vanes **168** can be incorporated into the engine **120**. The row **160** of blades **162** establishes a blade quantity (BQ). The row **166** of guide vanes **168** establishes a vane quantity (VQ), which can be the same or can differ from the blade quantity (BQ). The blade quantity (BQ) can be less than 24 or 26 blades, or more narrowly at least 12 but not more than 20 blades, such as 14 or 16 blades, for example. The vane quantity (VQ) can be no more than 40 guide vanes, or more narrowly at least 20 guide vanes, for example. In examples, the ratio of VQ/BQ is between 2.0 and 2.6, or more narrowly between 2.2 and 2.5, which may reduce noise due to flow interaction between the blades **162** and guide vanes **168**. For the purposes of this disclosure, the term “between” is inclusive of the stated value(s) unless otherwise stated.

(51) Various stage counts can be incorporated in the compressors **144**, **152** and turbines **146**, **154**. For example, the low pressure compressor **144** can include at least 3 but no more than 5 compressor stages, such as 3 compressor stages as illustrated by FIGS. **1** and **5**. In examples, the low pressure turbine **146** includes at least 3 but no more than 6 turbine stages, such as 3 turbine stages as illustrated by FIG. **5**, 4 turbine stages, or 5 turbine stages as illustrated by FIG. **1**. The high pressure turbine **154** can include a single stage, or can include two stages as illustrated by FIGS. **1** and **5**, for example. A ratio between the blade quantity (BQ) and a quantity of the stages of the low pressure turbine **146** can be between about 3.3 and about 8.6, or more narrowly less than or equal to about 4.0, such as about 3.5. The example low pressure turbine **146** provides the driving power to rotate the fan **142** and therefore the relationship between the number of turbine stages of the low pressure turbine **146** and the blade quantity (BQ) disclose an example gas turbine engine **120** with increased power transfer efficiency.

(52) The engine **120** includes a core bypass (or third) duct **127** established between the engine case **117** and core cowling **125**. The core bypass duct **127** establishes an intermediate (or third) flow path I. The core bypass duct **127** can be a separate and distinct structure or can be established by a volume of the core engine bay between the engine core and core cowling **125**. The core bypass duct **127** can be coupled to an annular core vent port (or nozzle) **129** established along an outer periphery of the core cowling **125**. In the illustrated example of FIG. **5**, the guide vanes **168** are an axially aftmost row of fan exit guide vanes in the bypass flow path B between the row **160** of blades **162** and the bypass exit **131** relative to the engine axis A. In other examples, one or more rows of vanes are arranged in the bypass flow path B axially forward and/or aft of the row **166** of guide vanes **168**. The hub **164** is rotatable about the engine axis A such that the row **160** of blades **162** deliver or convey airflow to the bypass, intermediate and core flow paths B, I, C in operation.

(53) The engine **120** includes a first flow splitter **170** and a second flow splitter **172** arranged in a cascade. The flow splitters **170**, **172** are situated downstream and axially aft of the row **160** of blades **162** relative to the engine axis A. The first flow splitter **170** can be incorporated into or mechanically attached to the core cowling **125**. The second flow splitter **172** can be incorporated into or mechanically attached to the engine case **117**. The first flow splitter **170** establishes a portion of the bypass duct **119** and a portion of the second duct **121**. The second flow splitter **172** establishes a portion of the second duct **121** and a portion of an entrance (e.g., fourth) duct **133**. The entrance duct **133** can establish an entrance or inlet to the compressor section **124**.

(54) The first flow splitter **170** is arranged to divide airflow conveyed by the row **160** of blades **162** between the bypass flow path B and the second duct **121**. The second flow splitter **172** is radially inboard of the first flow splitter **170** relative to the engine axis A. The second duct **121** branches between the core flow path C and the intermediate flow path I at the second flow splitter **172**. The

second flow splitter **172** is arranged to divide airflow conveyed to the second duct **121** between the intermediate flow path I and the core flow path C along the entrance duct **133**. The flow splitters **170**, **172** can have a generally V-shaped cross sectional geometry dimensioned to divide the airflow.

(55) A bypass port **174** is established along an outer periphery **125P** of the core cowling **125**. The bypass port **174** interconnects the intermediate flow path I and bypass flow path B at a position downstream of the row **166** of guide vanes **168** relative to a general direction of airflow through the bypass flow path B such that airflow conveyed by the intermediate flow path I through the bypass port **174** bypasses or otherwise is not communicated across the guide vanes **168** in operation. The bypass port **174** can interconnect the intermediate flow path I and bypass flow path B at a position axially aft of the 0% span position of the row **166** of guide vanes **168** relative to the engine axis A. In the illustrated example of FIG. 5, the vent port **129** is established downstream and axially aft of both the bypass port **174** and the bypass exit **131** relative to the engine axis A. In other examples, the vent port **129** is established upstream and axially forward of the bypass exit **131** such that airflow in the bypass flow path B mixes in the bypass duct **119** with airflow conveyed by the vent port **129** prior to exiting the bypass exit **131**.

(56) One or more louvers **176** can be situated in the bypass port **174**. The louvers **176** can have a generally airfoil-shaped cross sectional geometry and are arranged to direct airflow from the intermediate flow path I into the bypass flow path B in a direction generally downstream of the guide vanes **168** and toward the bypass exit **131**.

(57) The engine **120** can establish an adaptive fan flow arrangement in which a relative distribution of airflow conveyed by the fan **142** to the flow paths B, I and/or C varies during one or more modes. One or more blocker doors **184** can be moved between opened and closed positions to selectively communicate airflow through the bypass port **174** (open and closed positions indicated in dashed lines at **184**, **184'** of FIG. 8A for illustrative purposes).

(58) In the illustrative example of FIG. 6, the engine **120** includes one or more blocker doors **178** situated in the core bypass duct **127** to modulate airflow through the intermediate flow path I. The blocker doors **178** are situated in the intermediate flow path I downstream of the bypass port **174**. The blocker doors **178** are moveable between a closed position (FIG. 5) and an open position (FIG. 6) to selectively communicate airflow in the intermediate flow path I downstream to the vent port **129**. The blocker doors **178** at least partially or substantially block airflow through the intermediate flow path I in the closed position. In the illustrative example of FIGS. 5-6, the blocker doors **178** are pivotable flaps having a pivot point established along the core bypass duct **127**. It should be understood that other arrangements can be utilized to selectively communicate airflow downstream to the vent port **129**.

(59) Referring to FIGS. 7A-7C, with continuing reference to FIGS. 5-6, the nacelle **123** can include at least one thrust reverser **180** and/or a variable area fan nozzle (VAFN) **182** for adjusting various characteristics of the bypass flow. FIG. 7A illustrates the thrust reverser **180** and the variable area fan nozzle **182** in stowed positions. FIG. 7B illustrates the thrust reverser **180** in the stowed position and the variable area fan nozzle **182** in a deployed position. FIG. 7C illustrates the thrust reverser **180** in a deployed position and the variable area fan nozzle **182** in the stowed position. The nacelle **123** includes a stationary portion **123-1** mounted to the fan case **115**, a first moveable portion **123-2** and a second movable portion **123-3**. The moveable portions **123-2**, **123-3** are moveable relative to the stationary portion **123-1** and relative to each other.

(60) The thrust reverser **180** is operable to convey airflow in the bypass flow path B for producing reverse thrust such as during approach and/or landing conditions of an aircraft associated with the engine **120**. The thrust reverser **180** includes a thrust reverser body **180B**, which is configured with the first moveable portion **123-2** of the nacelle **123**. The thrust reverser body **180B** may have a generally tubular geometry. The thrust reverser **180** can include one or more blocker doors **180D**, one or more actuators **180A**, and/or one or more cascades **180C** of turning vanes **180V** arranged

circumferentially about the engine axis A.

(61) The thrust reverser body **180B** can include at least one recess **180R** that houses the cascades **180C** and the actuators **180A** when the thrust reverser **180** is in the stowed position. The cascades **180C** are dimensioned to span across a reversal port **180P** when the thrust reverser **180** is in the deployed position, as illustrated by FIG. 7C. The cascade **180C** selectively communicates airflow from the bypass duct **119** outwardly through the reversal port **180P** in the deployed position. The cascade **180C** can extend axially aft of both the row **166** of guide vanes **168** and the bypass port **174** with respect to the engine axis A, and the row **168** of guide vanes **168** can be situated axially forward of the blocker doors **180D** relative to the engine axis A, as illustrated by FIGS. 5-6.

(62) Each blocker door **180D** is pivotally connected to the thrust reverser body **180B**. The bypass port **174** can be established axially forward of the blocker doors **180D** with respect to the engine axis A, as illustrated in FIGS. 5-6. The actuators **180A** are operable to axially translate the thrust reverser body **180B** between the stowed and deployed positions. As the thrust reverser body **180B** translates aftwards, the blocker doors **180B** pivot radially inward into the bypass duct **119** and divert at least some or substantially all of the bypass airflow F as flow F1 through the cascades **180C** to provide the reverse engine thrust, as illustrated in FIG. 7C. In other examples, the cascades **180C** are configured to translate axially with a respective thrust reverser body **180B**. The thrust reverser body **180B** and/or cascades **180C** can include one or more circumferential segments that synchronously or independently translate or otherwise move between deployed and stowed positions.

(63) The variable area fan nozzle **182** includes a nozzle body **182B** and one or more actuators **182A**. The nozzle body **182B** is configured with the second movable portion **123-3** of the nacelle **123**, and is arranged radially within and may nest with the thrust reverser body **180B**. The nozzle body **182B** may have a generally tubular geometry that extends about an axially contoured outer periphery **125P** of the core cowling **125**. The actuators **182A** are operable to axially translate the nozzle body **182B** between the stowed position of FIG. 7A and the deployed position of FIG. 7B. As the nozzle body **182B** translates aftwards, a radial distance RD of the bypass exit **131** between a trailing edge or aft end **123TE** of the nacelle **123** and the outer periphery **125P** of the core cowling **125** established at the stowed position may change (e.g., increase) to a radial distance RD' and thereby change (e.g., increase) a flow area of the bypass exit **131**. In this manner, the variable area fan nozzle **182** may adjust a pressure drop or ratio across the bypass flowpath B defined by the bypass duct **119** by changing the flow area of the bypass exit **131**.

(64) The variable area fan nozzle **182** can define or otherwise include at least one auxiliary port **182P** to vary the bypass flow. The auxiliary port **182P** can be established in the nozzle body **182B**, as illustrated by FIGS. 7A-7C. The auxiliary port **182P** is moveable relative to a trailing edge portion **123P** of the first moveable portion **123-1** of the nacelle **123** to selectively block and open the auxiliary port **182P**. The auxiliary port **182P** is opened as the nozzle body **182B** translates axially aftwards relative to the engine axis A. A flow area through the auxiliary port **182P** augments the flow area of the bypass exit **131**, thereby increasing an effective flow area of the variable area fan nozzle **182**. The variable area fan nozzle **182** therefore may adjust a pressure drop or ratio across the bypass flowpath B defined by the bypass duct **119** while translating the nozzle body **182B** over a relatively smaller axial distance. In the illustrative example of FIGS. 5-6, the variable area fan nozzle **182** is omitted.

(65) Referring to FIG. 8A, with continuing reference to FIG. 5, the blades **162** can be dimensioned to provide a relatively radially compact engine arrangement. Each of the blades **162** extends in the radial direction R between a 0% span position at the hub **164** and a 100% span position at a tip **162T**. A tip radius RT of the blades **162** is established between a radially outermost portion of the tip **162T** and the engine axis A. A forwardmost portion of a leading edge **162LE** of each the blades **162** is arranged along a first reference plane REF1. The first reference plane REF1 is perpendicular to the engine axis A. A hub radius RH is established along the first reference plane REF1 between

an outer periphery of the hub **164** and the engine axis A. The tip radius RT can be between 35 and 60 inches, or more narrowly between 40 and 55 inches, for example.

(66) Each of the blades **162** extends in the radial direction R between the 0% span position and the 100% span position to establish a geometric hub-to-tip ratio (RH:RT). The hub-to-tip ratio (RH:RT) can be less than or equal to about 0.4, or more narrowly greater than or equal to about 0.1, measured relative to the forwardmost portion of the leading edge **162LE** of the blades **162**. In examples, the hub-to-tip ratio (RH:RT) is between 0.16-0.36, or more narrowly between 0.2-0.3. In examples, the hub-to-tip ratio (RH:RT) is established such that a maximum value of the fan pressure ratio across the blade **162** alone is less than or about equal to about 1.45 measured at cruise at 0.8 Mach and 35,000 feet. In further examples, a maximum value of the fan pressure ratio is greater than or equal to about 1.1, or more narrowly less than or equal to about 1.35. The fan pressure ratios and hub-to-tip ratios disclosed herein can be utilized alone or in combination to establish relatively high bypass turbo fan engine arrangements.

(67) The flow splitters **170**, **172** are arranged at respective positions relative to each other and the row **160** of blades **162** to establish a predetermined distribution of airflow through the bypass, intermediate and core flow paths B, I, C in operation. A forwardmost edge **170F** of the first flow splitter **170** is situated in the radial and axial directions R, H at a first splitter position P1 relative to the engine axis A. The first splitter position P1 is established along a second reference plane REF2. A forwardmost edge **172F** of the second flow splitter **172** is situated in the radial and axial directions R, H at a second splitter position P2 relative to the engine axis A. The second splitter position P2 is established along a third reference plane REF3. The reference planes REF2, REF3 are perpendicular to the engine axis A.

(68) The reference planes REF2, REF3 can be established at the same position, or at different positions relative to the engine axis A to establish a cascade arrangement. For example, the forwardmost edge **172F** of the second flow splitter **172** can be axially aft of the forwardmost edge **170F** of the first flow splitter **170** with respect to the engine axis A. The forwardmost edges **170F**, **172F** of the flow splitters **170**, **172** are axially forward of the leading edges **168LE** of the row **166** of guide vanes **168** at the 0% span position with respect to the engine axis A. The forwardmost edges **170F**, **172F** can be dimensioned to extend axially forward of the geared architecture **148** relative to the engine axis A, as illustrated by FIG. 8A, which can establish a relatively compact core engine arrangement.

(69) Referring to FIG. 8B, with continuing reference to FIG. 8A, a first annulus area A1 is established by, and between, inner and outer diameter surfaces **119A**, **119B** of the bypass duct **119** at the forwardmost edge **170F** of the first flow splitter **170**. A second annulus area A2 is established by, and between, inner and outer diameter surfaces **121A**, **121B** of the second duct **121** at the forwardmost edge **170F** of the first flow splitter **170**. A third annulus area A3 is established by, and between, inner and outer diameter surfaces **133A**, **133B** of the entrance duct **133** at the second flow splitter **172**. The annulus areas A1, A2, A3 extend about the engine axis A to establish a portion of the respective flow paths B, I, C.

(70) The annulus areas A1, A2, A3 can be dimensioned to establish a predetermined distribution of airflow to the flow paths B, I, C in operation. A first geometric bypass area ratio (A1:A2) associated with the first flow splitter **170** is defined as the first annulus area A1 divided by the second annulus area A2. A second geometric bypass area ratio (A1:A3) associated with the second flow splitter **172** is defined as the first annulus area A1 divided by the third annulus area A3. The annulus areas A1, A2, A3 can be dimensioned to establish a predetermined value of the first bypass area ratio (A1:A2) and a predetermined value of the second bypass area ratio (A1:A3). In examples, the first bypass area ratio (A1:A2) is greater than or equal to about 10 to establish a relatively high bypass turbo fan arrangement. In examples, the first bypass area ratio (A1:A2) is less than or equal to about 35, or more narrowly less than or equal to about 20. The second bypass area ratio (A1:A3) is greater than the first bypass area ratio (A1:A2). For example, a value of the

second bypass area ratio ($A1:A3$) can be between about 150-250% of the first bypass area ratio ($A1:A2$), such as approximately twice or 200% of the first bypass area ratio ($A1:A2$). In examples, the second bypass area ratio ($A1:A3$) is greater than or equal to about 15, or more narrowly greater than or equal to about 20.

(71) Referring back to FIG. 8A, the forwardmost edges **170F**, **172F** of the flow splitters **170**, **172** can be dimensioned relative to a span of the blades **162** to establish the predetermined distribution of airflow through the bypass, intermediate and core flow paths B, I, C in operation. In examples, the first splitter position **P1** can be radially aligned with or radially inward of a 25% span position of the blades **162**, or more narrowly can be radially aligned with or radially outward of a 10% span position of the blades **162** relative to the radial direction R, such as between about 16-22% span. The second splitter position **P2** can be radially aligned with or radially outward of a 5% span position of the blades **162**, or more narrowly can be radially aligned with or radially inward of a 15% span position of the blades **162** relative to the radial direction R, such as between about 8-13% span. The splitter positions **P1**, **P2** can be established at a radial distance of at least 5% span apart from each other relative to the blades **162** such that a relatively larger amount of airflow is communicated to the intermediate flow path I and bypasses the guide vanes **168**. In examples, the radial distance between the splitter positions **P1**, **P2** is between about 6-12% span, relative to the blades **162**. The blades **162** can have any of the stagger angles α and/or camber κ values disclosed herein at the respective spanwise positions of the first splitter position **P1** and/or second splitter position **P2** to convey airflow in a predetermined direction towards the splitters **170**, **172**, which can improve efficiency.

(72) The guide vanes **168** and flow splitters **170**, **172** can be arranged relative to the fan blades **162** to establish a relatively axially compact envelope. The leading edges **162LE** of the blades **162** at the 100% span position are established along a fourth reference plane REF4. The leading edges **168LE** of the guide vanes **168** at the 100% span position are established along a fifth reference plane REF5. The reference planes REF4, REF5 are perpendicular to the engine axis A. A first axial length L1 is established between the reference planes REF4, REF5. In examples, a ratio ($L1:RT$) of the first axial length L1 divided by the tip radius RT is greater than or equal to about 0.5, which can reduce fan noise propagation through the bypass flow path B. The ratio ($L1:RT$) can be less than or equal to about 0.75, which can establish a relatively axially compact fan section. A second axial length L2 is established between the first and second splitter positions **P1**, **P2** along the reference planes REF2, REF3 relative to the engine axis A. In examples, a ratio ($L2:L1$) of the second axial length L2 divided by the first axial length L1 is less than or equal to about 0.5, or more narrowly between about 0.1 and about 0.3, which can further establish a relatively axially compact engine.

(73) FIG. 9A illustrates a plot of bypass area ratio relative to airfoil span position, such as span positions of the blades **162**. FIG. 9B illustrates selected portions of FIG. 9A. Example profiles or curves C1-C6 are shown. The curves C1-C6 correspond to fan or rotor arrangements having hub-to-tip ratios ($RH:RT$) of 0.16, 0.20, 0.24, 0.28, 0.32, 0.36, respectively. The fan **142** can be dimensioned to establish any of the hub-to-tip ratios ($RH:RT$) and corresponding values along the curves C1-C6.

(74) Each curve C1-C6 can be established as function of annulus area associated with respective bypass area ratios. Splitter position values for each % span position along the respective curve C1-C6 can be defined in one of the following ways:

$$\% \text{ span position} = [(MR - RH) / (RT - RH)] * 100 \quad \text{Equation 1}$$

$$\% \text{ span position} = [(MR/RT - \lambda) / (1 - \lambda)] * 100 \quad \text{Equation 2}$$

where (MR) is the radial splitter position of the respective splitter **170**, **172**, (RH) is the hub radius and (RT) is the tip radius as previously defined, and (λ) is the hub-to-tip ratio ($RH:RT$). Since the annulus areas established by the splitters are non-linear, the corresponding curves C1-C6 are also non-linear.

(75) The plot of curves C1-C6 can serve as a nomograph utilized to select radial locations of the

first and second positions **P1**, **P2** of the flow splitters **170**, **172** with respect to hub-to-tip (RH:RT) ratio and/or bypass area ratio (**A1:A2**), (**A1:A3**), or vice versa. For example, referring to FIG. **9B**, the fan **142** can establish a hub-to-tip (RH:RT) ratio of 0.16 corresponding to the curve **C1**. The radial location of the first and second splitter positions **P1**, **P2** of the flow splitters **170**, **172** correspond to points **P1.sub.(0.16)**, **P2.sub.(0.16)** selected along the curve **C1**. The points **P1.sub.(0.16)**, **P2.sub.(0.16)** can be selected with respect to a predetermined bypass area ratio. For example, point **P1.sub.(0.16)** of the first flow splitter **170** can be established in the radial direction **R** at about a 21.5% span position of the blades **162**. Point **P1.sub.(0.16)** corresponds to a first bypass area ratio (**A1:A2**) of about 10 and a mass average fan pressure ratio of about 1.260. The radial location of the second flow splitter **172** can be determined by assuming analytically that the second flow splitter **172** is the first flow splitter **170** (e.g., a single flow splitter arrangement). The point **P2.sub.(0.16)** of the second flow splitter **172** can be established in the radial direction **R** at about a 13.1% span position of the blades **162**. Point **P2.sub.(0.16)** corresponds to a second bypass area ratio (**A1:A3**) of about 20 and a mass average fan pressure ratio of about 1.225. For the purposes of this disclosure, mass average fan pressure ratio is defined as an average pressure ratio across the blades **162** alone measured at cruise at 0.8 Mach and 35,000 feet between the 0% span position and the corresponding span position of the respective splitter **170**, **172**. The mass across the blades **162** associated with the mass average fan pressure ratio is approximately equal to the amount of the mass captured by the respective splitter **170**, **172**.

(76) As another example, the fan **142** can establish a hub-to-tip (RH:RT) ratio of 0.36 corresponding to the curve **C6**. The radial location of the first and second splitter positions **P1**, **P2** of the flow splitters **170**, **172** correspond to points **P1.sub.(0.36)**, **P2.sub.(0.36)** selected along the curve **C6**. Point **P1.sub.(0.36)** of the first flow splitter **170** can be established in the radial direction **R** at about a 15.5% span position of the blades **162**. Point **P1.sub.(0.36)** corresponds to a first bypass area ratio (**A1:A2**) of about 10 and a mass average fan pressure ratio of about 1.260. Point **P2.sub.(0.36)** of the second flow splitter **172** can be established in the radial direction **R** at about a 7.6% span position of the blades **162**. Point **P2.sub.(0.36)** corresponds to a second bypass area ratio (**A1:A3**) of about 20 and a mass average fan pressure ratio of about 1.185.

(77) The flow splitters **170**, **172** and other portions of the fan section **122** can be dimensioned with respect to any other values along the curves **C1-C6**. For example, points **P1'.sub.(0.16)**, **P2'.sub.(0.16)** can be established along the curve **C1** at about 11.0% and about 7.6% span corresponding to first and second bypass area ratios (**A1:A2**), (**A1:A3**) of about 25 and 35, respectively. As another example, points **P1'.sub.(0.36)**, **P2'.sub.(0.36)** can be established along the curve **C6** at about 7.0% and about 5.5% span corresponding to first and second bypass area ratios (**A1:A2**), (**A1:A3**) of about 20 and 35, respectively.

(78) As illustrated by the curves **C1-C6** of FIGS. **9B**, for the same bypass area ratio, a lower hub-to-tip ratio (RH:RT) can result in the splitter positions **P1**, **P2** of the splitters **170**, **172** being relatively further away from the 0% span position of the blades **162**. The splitters **170**, **172** can be dimensioned with respect to relatively lower % span positions to establish relatively larger bypass area ratios, as illustrated by the curves **C1-C6** of FIG. **9B**.

(79) FIG. **10A** illustrates a plot of fan pressure ratio relative to airfoil span position, such as span positions of the blades **162**. FIG. **10B** illustrates selected portions of the plot of FIG. **10A**. Curve **PR1** corresponds to discrete fan pressure ratios at each span position. Lines **LP1.sub.(0.16)**, **LP2.sub.(0.16)** correspond to the splitter positions **P1.sub.(0.16)**, **P2.sub.(0.16)** of the splitters **170**, **172** associated with a hub-to-tip ratio (RH:RT) equal to 0.16. Lines **MA1.sub.(0.16)**, **MA2.sub.(0.16)** are the mass averaged fan pressure ratios corresponding to the respective splitter positions **P1.sub.(0.16)**, **P2.sub.(0.16)**. FIG. **10B** depicts a curve **PR1(NT)** representing a spanwise integration of the discrete fan pressure ratios of curve **PR1**. An intersection of the lines **LP1.sub.(0.16)**, **LP2.sub.(0.16)** corresponding to the points **P1.sub.(0.16)**, **P2.sub.(0.16)** and the curve **PR1(INT)** establish a position of the respective lines **MA1.sub.(0.16)**, **MA2.sub.(0.16)** relative to

the y-axis. In the illustrative example of FIGS. **10A-10B**, lines **MA1.sub.(0.16)**, **MA2.sub.(0.16)** correspond to mass average fan pressure ratios of 1.260 and 1.225, respectively, which correspond to values previously discussed with respect to the points **P1.sub.(0.16)**, **P2.sub.(0.16)** along the curve **C1** of FIG. **9B**.

(80) FIG. **11** illustrates another plot of fan pressure ratio relative to airfoil span position, such as span positions of the blades **162**. Curve **PR2** corresponds to discrete fan pressure ratios at each span position of the blade **162**. Lines **LP1.sub.(0.36)**, **LP2.sub.(0.36)** correspond to splitter positions **P1.sub.(0.36)**, **P2.sub.(0.36)** of the splitters **170**, **172** associated with a hub-to-tip ratio (**RH:RT**) equal to 0.36. In the illustrative example of FIG. **11**, lines **MA1.sub.(0.36)**, **MA2.sub.(0.36)** correspond to mass average fan pressure ratios of 1.260 and 1.185, respectively, which correspond to values previously discussed with respect to the points **P1.sub.(0.36)**, **P2.sub.(0.36)** along the curve **C6** of FIG. **9B**.

(81) The engine **120** can include a controller **CONT** (shown in dashed lines in FIG. **5** for illustrative purposes). The controller **CONT** is operable to cause the blocker doors **178**, **184**, thrust reverser **180** and variable area fan nozzle **182** to move between the various disclosed positions or modes. The controller **CONT** can include one or more computing devices, each having one or more of a computer processor, memory, storage means, network device and input and/or output devices and/or interfaces according to some example. The memory may, for example, include **UVROM**, **EEPROM**, **FLASH**, **RAM**, **ROM**, **DVD**, **CD**, a hard drive, or other computer readable medium which may store data and/or the algorithms corresponding to the various functions of this disclosure. In other examples, the controller **CONT** is an analog or electromechanical device configured to provide the disclosed functions of this disclosure. The controller **CONT** can be a portion of a full-authority digital electronic control **FADEC** or an electronic engine control (**EEC**), another system, or a stand-alone system located within the aircraft remote from the engine **120**.

(82) FIG. **12** illustrates a plurality of discrete modes **M1-M4** established by the position of the blocker door(s) **178** and position of the blocker door(s) **184** across the bypass port **174**. The controller **CONT** can be programmed with logic or otherwise configured to cause the engine **120** to execute each of the modes **M1-M4**. During modes **M1** and **M3**, the blocker doors **178** are stowed to establish an open position to at least partially communicate airflow through vent port **129**. During modes **M2** and **M4**, the blocker doors **178** are deployed to establish a closed position to at least partially block airflow from being communicated to the vent port **129**. During modes **M1** and **M2**, the bypass port **174** is open or uncovered to at least partially communicate airflow from the intermediate flow path **I** to the bypass flow path **B**. During modes **M3** and **M4**, the bypass port **174** is closed or covered to at least partially block airflow being communicated from the intermediate flow path **I** to the bypass flow path **B**. The engine **120** can be operated in the first mode **M1** during takeoff, for example, which may be associated with relatively lower engine inlet airflow such that the fan **142** communicates relatively more airflow which may improve fan stability.

(83) The engine **120** can be operated in the first mode **M1** and/or the second mode **M2** during cruise, for example. The engine **120** can be operated in the second mode **M2** during a beginning of climb condition to increase fan pressure ratio and generate relatively greater thrust. The thrust reverser **180** can be deployed during the second mode **M2**, for example, to increase an amount of airflow being communicated to the bypass flow path to the thrust reverser **180** for generating reverse thrust. The engine **120** can be operated in the third mode **M3** during a middle of climb condition (i.e., between beginning climb and top of climb). The engine **120** can be operated in the fourth mode **M4** during a top of climb condition to increase airflow to the core flow path **C** and bypass flow path **B** to maximize or otherwise increase thrust. The modes **M1-M4** are exemplary and it should be appreciated the blocker door(s) **178** and bypass port **174** can be configured to establish fewer or more than four modes.

(84) The controller **CONT** can be coupled to one or more sensors to determine any of the conditions disclosed herein, such as temperature, pressure and/or speed sensors operable to

measure various conditions or states of the engine **120**. Temperature and/or pressure sensors can be situated along the bypass, intermediate and/or core flow paths B, I, C, for example. Speed sensors can be situated adjacent to the rotors to determine a speed of the fan rotor, low spool, and/or high spool, for example. The controller CONT can utilize other data and information to determine any of the conditions disclosed herein, including aircraft velocity, altitude and throttle position. One would understand how to situate the sensors and program or otherwise configure the controller CONT with logic to obtain data and other information from the sensors and other systems in view of the teachings disclosed herein.

(85) The engine **120** can be operated as follows. Referring to FIG. 5, the fan **142** is driven by the fan drive turbine **146** such that the blades **162** deliver airflow to the bypass, intermediate and core flow paths B, I, C. Driving the fan **142** can include driving the hub **164** of the fan **142** through the geared architecture **148** at a different speed than a speed of the fan drive turbine **146**. The engine **120** can be operated in any of the modes disclosed herein, including the modes M1-M4 of FIG. 12. Airflow in the intermediate flow path I can be ejected from the bypass port **174** into the bypass flow path B to bypass the guide vanes **168**. One or more blocker doors **178** can be selectively moved between open and closed positions to at least partially block airflow from the intermediate flow path I being communicated downstream to the core bypass duct **127** and vent port **129**. One or more of the blocker doors **184** can be moved between open and closed positions to selectively communicate airflow through the bypass port **174** (**184** shown in dashed lines in FIG. 8A for illustrative purposes).

(86) Referring to FIG. 6, an operating line of the fan **142** at an operating condition such as takeoff can be shifted relatively further away from the stall line by upflowing the fan **142** in response to selectively communicating airflow to the vent port **129**. One or more of the blocker doors **178** can be selectively moved to the open position to communicate airflow in the intermediate flow path I downstream to the vent port **129**. Airflow communicated to the core bypass duct **127** can be ejected from the vent port **129** downstream of the bypass port **174** and/or bypass exit **131**.

(87) The disclosed splitter arrangements can improve aerodynamic loading and efficiency of the guide vanes, including lowering inlet Mach number to the guide vanes and improving aerodynamic diffusion in the guide vanes. The disclosed splitter arrangements can improve blade and guide vane performance near 0% span, which can be utilized to achieve a relatively more efficient spanwise pressure ratio across the entire span of the blade and the entire span of the guide vane. Engines made with the disclosed architecture, and including arrangements as set forth in this application, and with modifications coming from the scope of the claims in this application, thus provide very high efficient operation, relatively high stall margins, and are compact and lightweight relative to their thrust capability. Two-spool and three-spool direct drive engine architectures can also benefit from the teachings herein.

(88) It should be understood that relative positional terms such as “forward,” “aft,” “upper,” “lower,” “above,” “below,” and the like are with reference to the normal operational attitude of the vehicle and should not be considered otherwise limiting.

(89) Although the different examples have the specific components shown in the illustrations, embodiments of this disclosure are not limited to those particular combinations. It is possible to use some of the components or features from one of the examples in combination with features or components from another one of the examples.

(90) Although particular step sequences are shown, described, and claimed, it should be understood that steps may be performed in any order, separated or combined unless otherwise indicated and will still benefit from the present disclosure.

(91) The foregoing description is exemplary rather than defined by the limitations within. Various non-limiting embodiments are disclosed herein, however, one of ordinary skill in the art would recognize that various modifications and variations in light of the above teachings will fall within the scope of the appended claims. It is therefore to be understood that within the scope of the

appended claims, the disclosure may be practiced other than as specifically described. For that reason the appended claims should be studied to determine true scope and content.

Claims

1. A section for a gas turbine engine comprising: a rotor including a row of blades extending in a radial direction outwardly from a hub, wherein the hub is rotatable about a longitudinal axis such that the row of blades deliver flow to a bypass flow path, an intermediate flow path and a core flow path; a first case surrounding the row of blades to establish the bypass flow path through a first duct; a first flow splitter that divides flow between the bypass flow path and a second duct; an aftmost row of guide vanes extending in the radial direction across the bypass flow path; a second case including a second flow splitter radially inboard of the first flow splitter, wherein the second flow splitter is axially forward of an axially forwardmost compressor blade row of the gas turbine engine relative to the longitudinal axis that is surrounded by the second case, the second flow splitter divides flow from the second duct between the intermediate flow path and the core flow path, a forwardmost edge of the second flow splitter is axially aft of a forwardmost edge of the first flow splitter with respect to the longitudinal axis, and the forwardmost edges of the first and second flow splitters are axially forward of the aftmost row of guide vanes with respect to the longitudinal axis; a bypass port that interconnects the intermediate and bypass flow paths at a position axially aft of a position of the aftmost row of guide vanes relative to the longitudinal axis, and the bypass port configured to direct flow from the intermediate flow path to the bypass flow path; wherein each of the blades extends in the radial direction between a 0% span position at the hub and a 100% span position at a tip, leading edges of the blades at the 100% span position are established along a first reference plane, leading edges of the guide vanes at a 100% span position of the respective guide vanes are established along a second reference plane, the first and second reference planes are perpendicular to the longitudinal axis, a first axial length is established between the first and second reference planes, a tip radius is established between the tip of one of the respective blades and the longitudinal axis, and a ratio of the first axial length divided by the tip radius is greater than or equal to 0.5; and wherein the forwardmost edge of the first flow splitter is situated in the radial direction at a first splitter position, the forwardmost edge of the second flow splitter is situated in the radial direction at a second splitter position, the first splitter position is radially aligned with or radially inward of a 25% span position of the blades, and the second splitter position being radially aligned with or radially outward of a 5% span position of the blades.

2. The section as recited in claim 1, wherein the hub is driven by a turbine through a geared architecture.

3. The section as recited in claim 1, wherein each of the blades extends in the radial direction between the 0% span position at the hub and the 100% span position at the tip to establish a hub-to-tip ratio, and the hub-to-tip ratio of each one of the blades is less than or equal to 0.4, the hub-to-tip ratio of each one of the blades measured relative to the forwardmost portion of the leading edge of the respective blade.

4. The section as recited in claim 1, wherein the first duct establishes the bypass flow path, the second duct branches between the intermediate and core flow paths at the second flow splitter, a first annulus area is established by the first duct at the forwardmost edge of the first flow splitter, a second annulus area is established by the second duct at the forwardmost edge of the first flow splitter, a bypass area ratio is defined as the first annulus area divided by the second annulus area, and the bypass area ratio is greater than or equal to 10, and less than or equal to 35.

5. The section as recited in claim 4, wherein a maximum value of a pressure ratio across the blades alone is less than 1.45 at cruise at 0.8 Mach and 35,000 feet.

6. The section as recited in claim 4, wherein only one row of guide vanes extends in the radial direction across the bypass flow path and the row of guide vanes comprises the aftmost row of

guide vanes.

7. The section as recited in claim 4, wherein each of the blades extends in the radial direction between the 0% span position at the hub and the 100% span position at the tip to establish a hub-to-tip ratio, and the hub-to-tip ratio of each one of the respective blades is between 0.16-0.36 measured relative to the forwardmost portion of the leading edge of the respective blade.

8. The section as recited in claim 7, wherein the row of blades establishes a blade quantity (BQ), the blade quantity (BQ) is at least 12 but not more than 20 blades, the aftmost row of guide vanes establishes a vane quantity (VQ), and the vane quantity (VQ) is at least 20 but not more than 40 guide vanes.

9. The section as set forth in claim 8, wherein a ratio of VQ/BQ is between 2.0 and 2.6.

10. The section as recited in claim 1, further comprising: one or more blocker doors situated in the intermediate flow path downstream of the bypass port, and wherein the one or more blocker doors are moveable to direct flow from the intermediate flow path to the bypass port.

11. The section as set forth in claim 10, wherein: the one or more blocker doors are configured to pivot from the second case towards the bypass flow path to direct flow from the intermediate flow path to the bypass port; and louvers are situated in the bypass port, the louvers have an airfoil-shaped geometry and are arranged to direct flow from the intermediate flow path to the bypass flow path.

12. A gas turbine engine comprising: a fan section including a fan having a row of blades, each extending in a radial direction between a 0% span position at a hub and a 100% span position at a respective tip, wherein the hub is rotatable about an engine longitudinal axis such that the row of blades deliver flow to a bypass flow path, an intermediate flow path and a core flow path; a compressor section establishing the core flow path; a turbine section that drives the fan section and the compressor section; a fan case including a bypass duct surrounding the row of blades to establish the bypass flow path; a housing including a first flow splitter that divides flow between the bypass flow path and a second duct; an aftmost row of guide vanes in the bypass duct that extend in the radial direction across the bypass flow path; an engine case including a second flow splitter radially inboard of the first flow splitter and that divides flow from the second duct between the intermediate flow path and the core flow path, wherein the second flow splitter is axially forward of an axially forwardmost compressor blade row of the compressor section that is surrounded by the engine case relative to the engine longitudinal axis, a forwardmost edge of the second flow splitter is axially aft of a forwardmost edge of the first flow splitter with respect to the engine longitudinal axis, and the forwardmost edges of the first and second flow splitters are axially forward of the row of aftmost guide vanes with respect to the engine longitudinal axis; a bypass port that interconnects the intermediate and bypass flow paths at a position downstream of the aftmost row of guide vanes, and the bypass port configured to direct flow from the intermediate flow path to the bypass flow path; wherein each of the blades extends in the radial direction between the 0% span position and 100% span position to establish a hub-to-tip ratio, and the hub-to-tip ratio of each one of the blades is less than or equal to 0.4 measured relative to a forwardmost portion of a leading edge of the respective blade; the leading edges of the blades at the 100% span position are established along a first reference plane, leading edges of the guide vanes at a 100% span position of the respective guide vanes are established along a second reference plane, the first and second reference planes are perpendicular to the engine longitudinal axis, a first axial length is established between the first and second reference planes, a tip radius is established between the tip of one of the respective blades and the longitudinal axis, and a ratio of the first axial length divided by the tip radius is greater than or equal to 0.5; and wherein the forwardmost edge of the first flow splitter is situated in the radial direction at a first splitter position, the forwardmost edge of the second flow splitter is situated in the radial direction at a second splitter position, the first splitter position is radially aligned with or radially inward of a 25% span position of the blades relative to the engine longitudinal axis, and the second splitter position is radially aligned with or radially

outward of a 5% span position of the blades relative to the engine longitudinal axis.

13. The gas turbine engine as recited in claim 12, further comprising a geared architecture, wherein the turbine section includes a fan drive turbine, and the geared architecture drives the fan at a different speed than a speed of the fan drive turbine.

14. The gas turbine engine as recited in claim 12, further comprising one or more blocker doors situated in the intermediate flow path downstream of the bypass port, wherein the one or more blocker doors are moveable between a first position and a second position, wherein the one or more blocker doors are configured to communicate flow in the intermediate flow path to a vent port in the first position but are configured to block flow in the intermediate flow path to the vent port in the second position.

15. The gas turbine engine as recited in claim 14, wherein the vent port is axially aft of an exit of the bypass flow path with respect to the engine longitudinal axis.

16. The gas turbine engine as recited in claim 14, further comprising a thrust reverser including a cascade that selectively communicates airflow from the bypass duct, wherein the cascade extends axially aft of both the aftmost row of guide vanes and the bypass port with respect to the engine longitudinal axis.

17. The gas turbine engine as recited in claim 12, wherein: the compressor section includes a first compressor and a second compressor, the first compressor comprises the axially forwardmost compressor blade row of the compressor section, and both the second flow splitter and an entrance of the intermediate flow path are axially forward of the compressor section relative to the engine longitudinal axis.

18. The gas turbine engine as recited in claim 17, wherein the first compressor includes three compressor blade rows.

19. The gas turbine engine as recited in claim 17, wherein the aftmost row of guide vanes extend in the radial direction across the intermediate flow path relative to the engine longitudinal axis.
